7CS039 Statistics for AI & Data Science

Week 4 Hypothesis Testing

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1 Introduction

The idea behind all statistical hypothesis testing is as follows:

- We have a theory (hypothesis) which we believe to be true (called the alternative hypothesis H_1).
- We only accept that it is true if we can show that the opposite (null hypothesis H_0) is very unlikely to be true.

How unlikely H_0 need be before we reject it is largely a matter for the researcher to determine, but the standard is that we only reject H_0, and accept H_1 if the probability that we would obtain a sample like that observed if the null hypothesis were true is at most 0.05. This is known as a *Level of Significance* of 0.05, we often write this as $\alpha = 0.05$. Sometimes if we wish to be extra sure that we do not come to the conclusion that our alternative hypothesis is true, when in fact it is false, (i.e. commit a type one error), we set a level of significance of 0.02 or 0.01, or if we need not be so certain that our conclusions are correct, we might let $\alpha = 0.10$.

2 General Setup

Some of the most common type of hypotheses that are tested are

- the mean of a population is equal to some given value.
- the mean of a population is greater/less than a given value.
- the means of two different populations are different.

Let's look at a simple example to get a better understanding of the general setup.

Example

Suppose that we wish to determine the mean lifespan of a given brand of energy saving lightbulb. The manufacturer claims that the mean $\mu = 600$ hours. We don't believe this and we have a suspicion that the mean is actually less than $\mu = 600$,

Example ctd.

Our null and alternative hypothesis are as follows

- H_0 - the mean is as the manufacturer say $\mu = 600$ hours. (we want to reject this)
- H_1 - the mean less than the manufacturer claims i.e. $\mu < 600$ hours. (we want to accept this)

This is an example of a *one-tailed* hypothesis test, H_0 is false if the true value of μ lies to the left side (one side) of the hypothesized value of μ.

We decide to let $\alpha = 0.05$. Thus we are only going to conclude that our alternative hypothesis is true if, in some sense, we can show that the probability that $\mu = 600$ is less than 0.05.

Suppose now for a moment that the mean lifespan of one of these lightbulbs *is* 600 hours as the manufacturer claims. This implies that about half the lightbulbs will last longer than 600 hours, and that about half will last less than 600 hours, thus if we take a sample of lightbulbs and find that $\bar{x} = 599.9$ we cannot conclude that $H_0 : \mu = 600$ is false because half of the time we would get a *sample* mean less than the population mean anyhow, and a difference of 0.1 means nothing. On the other hand, if we take a sample of bulbs and find $\bar{x} = 590$ it is less likely that the mean of the population from which the sample is drawn is equal to 600, and if $\bar{x} = 500$ it is even less likely. The question is: how far away from 600 need $\bar{x}$ be before we can conclude that the probability that $H_0 : \mu = 600$ is true is less than 0.05? The answer to this question will depend on two things:

- The standard deviation, σ, of the population from which the sample was taken. The larger the standard deviation the more spread out the data will be. Hence if in this case $\sigma = 20$ a difference of, say, 10 between μ and $\bar{x}$ is more significant than it would be if $\sigma = 50$.
- The size of the sample: the larger the sample, the more accurate the results, thus a difference of, say, 10 between μ and $\bar{x}$ is more significant if the sample size is 400 than if the sample size is 100.

3 Z–TESTS

Suppose we were to take a theoretically infinite number of samples of size n from data with mean μ and standard deviation σ, and record the mean of each sample. Then, with very very few exceptions, we would find that the standard deviation of these sample means is itself normally distributed with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.This result is known as the *central limit theorem*. The standard deviation of the sample means is referred to as the *standard error* of the mean. Similarly, if the lower quartile of each sample was calculated and recorded, and the standard deviation of all the lower quartiles

was calculated, this would be the *standard error of the lower quartiles*, and similarly for any statistic.

3.1 Z–TEST FOR A SINGLE POPULATION MEAN

Theoretically we should only use the Z-test for a single population mean (also known as the *single sample Z-test*) if we know the standard deviation (σ) of the population from which we took the sample. In practice we may also use the the Z-test for a single population mean if the sample size is greater than about thirty as then the sample standard deviation is usually a good approximation for the the standard deviation of the data from which the sample was taken. In this case we substitute s for σ in the formula below. The Z-test for a single population mean works as follows:

Definition 3.1. Let

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- We reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu \neq \mu_0$ if $|z_{\text{obs}}| > z_{\text{crit}}$ (Two tailed test)
- We reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu < \mu_0$ if $z_{\text{obs}} < z_{\text{crit}}$ ((Lower) One tailed test)
- We reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu > \mu_0$ if $z_{\text{obs}} > z_{\text{crit}}$ ((Upper) One tailed test)

Where z_{crit} is the *critical value* of z,

The value of z_{crit} depends upon the level of significance, and whether the test is one or two tailed. The most frequently used values are summarized in Table 1 below.

Table 1: Critical Values of Z

ONE TAILED						TWO TAILED		
$\mathbf{H_1}$	α	$\mathbf{z}_{\text{crit}}$	$\mathbf{H_1}$	α	$\mathbf{z}_{\text{crit}}$	$\mathbf{H_1}$	α	$\mathbf{z}_{\text{crit}}$
$H_1 : \mu < \mu_0$	0.01	-2.33	$H_1 : \mu > \mu_0$	0.01	2.33	$H_1 : \mu \neq \mu_0$	0.01	2.58
$H_1 : \mu < \mu_0$	0.02	-2.05	$H_1 : \mu > \mu_0$	0.02	2.05	$H_1 : \mu \neq \mu_0$	0.02	2.33
$H_1 : \mu < \mu_0$	0.05	-1.64	$H_1 : \mu > \mu_0$	0.05	1.64	$H_1 : \mu \neq \mu_0$	0.05	1.96
$H_1 : \mu < \mu_0$	0.10	-1.28	$H_1 : \mu > \mu_0$	0.10	1.28	$H_1 : \mu \neq \mu_0$	0.10	1.64

Example

Suppose that we wish to test whether or not H_1: the mean height of Wolverhampton men is greater than 69 inches i.e $\mu > 69$. We take a sample of size $n = 60$ and find that the mean of this sample is $\bar{x} = 70$, with a standard deviation of $s = 2$ inches. Is this sufficient evidence to claim that H_1 is true $\alpha = 0.05$?

Example ctd.

We must test $H_0 : \mu = 69$, at $\alpha = 0.05$. Here

$$z_{\text{obs}} = \frac{70 - 69}{2/\sqrt{60}} = 3.87.$$

We don't know the value of σ so we can substitute $s = 2$ as the sample is large. Since $z_{\text{obs}} = 3.87 > 1.64 = z_{\text{crit}}$ we can reject H_0, and thus conclude that the evidence is sufficient to conclude that the mean height of Wolverhampton men is greater than 69 inches.

3.2 Z-TEST FOR THE DIFFERENCE BETWEEN TWO POPULATION MEANS

The Z-test for the difference between two population means, also known as the two-sample Z-test, is used to test

$$H_1 : \mu_1 \neq \mu_2 \quad \text{or} \quad H_1 : \mu_1 < \mu_2 \quad \text{or} \quad H_1 : \mu_1 > \mu_2$$

Where μ_1 and μ_2 are the means of two different populations. For the two sample Z–test

$$z_{\text{obs}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}$$

The values of z_{crit} are exactly the same as for the one sample Z-test, see Table 1. Theoretically we should only use the two–sample Z–test if the variances σ_1^2 and σ_2^2 of the populations from which both samples were drawn is known, but in practice we may use the test provided both samples are large, which is usually taken to mean greater than 30, in which case we substitute the population variances by the sample variances.

Example

Suppose that we wish to test H_1: the mean height of Wolverhampton women and Birmingham women differs. Thus the null hypothesis is $H_0 : \mu_1 = \mu_2$, (where μ_1 and μ_2 are the population mean heights of Wolverhampton and Birmingham women respectively). A sample of 50 Wolverhampton women is taken, and it is found that $\bar{x}_1 = 64.3$ inches, with $s_1 = 3$ inches. A sample of 60 Birmingham women is taken, and it is found that $\bar{x}_2 = 64.8$ inches, with $s_2 = 2$ inches. Can we reject the above null hypothesis at $\alpha = 0.05$?

Example ctd.

Here:

$$z_{\text{obs}} = \frac{64.3 - 64.8}{\sqrt{\frac{3^2}{50} + \frac{2^2}{60}}} = -1.01$$

Note that this is a two–tailed test at $\alpha = 0.05$, thus $z_{\text{crit}} = 1.96$. Thus: $|z_{\text{obs}}| = 1.01 \not> 1.96 = z_{\text{crit}}$ Hence there is not sufficient evidence to reject H_0, i.e., the difference of half an inch in the mean heights of the two samples of women is not sufficient a large enough difference to be able to conclude with 95% certainty that the mean heights of the populations from which the samples were drawn differs.

4 t–TESTS....WHEN σ IS UNKNOWN

To use a Z-test we should know the standard deviation of the *population* from which we draw our sample(s). If the sample size is large, i.e. approximately > 30 we may still use the Z-test as the standard deviation of the sample (s) will closely approximate the standard deviation (σ) of the data from which the sample was taken; what if the sample size is ≤ 30? We use a t–*test*. At first glance, the t and z test look very similar. We look at the "one sample t–test" first.

4.1 t-TEST FOR A SINGLE POPULATION MEAN

This time we have

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

There are many different t–distributions (i.e. no standard one) so it is impossible to publish complete tables of all of them. In practise we'll use `R` so no need to worry.

Example

A car sales person claims that a particular model of car gives a mean mileage that differs from 20 miles per gallon. To test this claim an experiment was carried out in which a random sample of 10 such cars were each run on one gallon of petrol. The results were as follows.

$$24, 23, 18, 22, 19, 19, 22, 18, 18, 22$$

Are the sales person's claims justified at $\alpha = 0.05$ if we are testing $H_0 : \mu = 20$ against

i $H_1 : \mu > 20$,

ii $H_1 : \mu \neq 20$?

Example ctd.

It is easily shown that $\bar{x} = 20.5$, $s = 2.32$. Here $n = 10$.
Thus: $t_{\text{obs}} = \frac{20.5-20.0}{2.32/\sqrt{10}} = 0.682$

i) From `R` we have $t_{\text{crit}} = 1.833$ thus

$$t_{\text{obs}} = 0.682 \not> 1.833 = t_{\text{crit}}$$

Hence we may not reject H_0.

ii) This is a two tailed test and from `R` we have $t_{\text{crit}} = 2.262$.

Hence as:

$$|t_{\text{obs}}| = 0.682 \not> 2.262 = t_{\text{crit}}$$

we do not reject H_0.

5 p-VALUES

In problems such as these we often accompany the calculated test statistic with the corresponding p-value. In general the p-value may be defined as follows.

Definition 5.1. Corresponding to an observed value of a test statistic the p-value is the lowest level of significance for which the null hypothesis could have been rejected.

Example (again)

A car sales person claims that a particular model of car gives a mean mileage that differs from 20 miles per gallon. To test this claim an experiment was carried out in which a random sample of 10 such cars were each run on one gallon of petrol. The results were as follows.

$$24, 23, 18, 22, 19, 19, 22, 18, 18, 22$$

Are the sales person's claims justified at $\alpha = 0.05$ if we are testing $H_0 : \mu = 20$ against

i $H_1 : \mu > 20$,

ii $H_1 : \mu \neq 20$?

Example (again) ctd.

Now we use R to calculate the p-values.

i $H_1 : \mu > 20$,

```
> mileage<-c(24,23,18,22,19,22,19,18,18,22)
> t.test(mileage,mu=20,alternative="greater")
```

```
> mileage<-c(24,23,18,22,19,22,19,18,18,22)
> t.test(mileage,mu=20,alternative="greater")

        One Sample t-test

data:  mileage
t = 0.68111, df = 9, p-value = 0.2565
alternative hypothesis: true mean is greater than 20
95 percent confidence interval:
 19.15433      Inf
sample estimates:
mean of x
     20.5
```

The p-value is 0.2565 which is greater than $\alpha = 0.05$ so we can't reject H_0.

ii $H_1 : \mu \neq 20$?

```
> t.test(mileage,mu=20,alternative="two.sided")
```

```
> t.test(mileage,mu=20,alternative="two.sided")

        One Sample t-test

data:  mileage
t = 0.68111, df = 9, p-value = 0.5129
alternative hypothesis: true mean is not equal to 20
95 percent confidence interval:
 18.83937 22.16063
sample estimates:
mean of x
     20.5
```

The p-value is 0.5129 which is greater than $\alpha = 0.05$ so we can't reject H_0.

In practise we will use p-values as often as we can.

6 Other Similar Tests

There are a variety of other tests similar to the above including

- The Independent Samples t-test.
- The Paired Samples t-test.
- Mann-Whitney test.

- Chi–Square Tests.

We will look at some of these in the lab session this week.